

Fail-closed reproducibility for finite quantum sequential growth: exact controls, independent replay, and resource boundaries

Kenichi Osaki

Independent researcher

ORCID: [0009-0003-9256-7089](https://orcid.org/0009-0003-9256-7089)

Version 0.4.2, 20 August 2026

Scope statement. This manuscript reports the control and methodology layer only. It does not claim candidate-model validity, a continuum limit, a complete finite occurrence-ON semantics lattice, or a final theory.

Abstract

We report a fail-closed reproducibility framework for finite quantum sequential-growth calculations. The framework separates candidate evidence from controls, binds claims to versioned machine-readable artifacts, requires independent replay before promotion, and records wall-time, memory, disk, and exact-enumeration limits as open resource boundaries. Existing bounded artifacts include a 5-seed, 1,024-trajectory-per-seed small- n regression; 40 deterministic fixture trajectories over two sampler profiles; eight four-dimensional Minkowski-control fixtures; and five synthetic supervisor modes. These are implementation checks, not production measurements. In a separate exact random-growth scaling review, primary and independent paths complete growth through cardinality 38 with identical trajectory and relation digests. The last fully enumerated source, $n = 37$, has 3,791 down-sets; an attempted step from source size $n = 38$ detects at least 4,097 down-sets and fails closed at the declared cap of 4,096. This is a resource-limit observation, not an impossibility theorem. The active candidate uses a nonlocal link/diamond/precursor weight for which no exact reduced-state factorisation is certified, so no candidate production evidence is reported. The result is a reproducible methods record with explicit nonclaims, not a physical verdict on the candidate model.

1 Background and objective

Sequential-growth models construct a causal set one element at a time. The classical framework combines covariance and a Bell-causality condition with a stochastic growth law[4]; complex and quantum variants introduce additional extension, consistency, or operator questions [6, 2, 5]. Numerical causal-set studies also make clear that finite-size evidence and continuum claims require different controls[1]. The present work does not propose a new physical dynamics. Its objective is narrower: to make the authorization boundary around finite computations explicit and reproducible.

The contribution is a concrete workflow with four linked parts:

1. exact arithmetic and deterministic ticket generation for bounded sampler fixtures;
2. independent implementations that replay the same frozen wire format;
3. external resource supervision with atomic, digest-checked promotion;
4. a verdict grammar that prevents finite tests and resource exhaustion from being promoted to physical or mathematical no-go claims.

All numerical counts reported below are read from committed result artifacts. No new candidate trajectory or solver calculation is introduced by manuscript assembly.

2 Scope and threat model

Finite exact calculations can produce convincing numerical output while still leaving the semantic scope, sampling measure, replay independence, and resource boundary underspecified. The purpose of this work is to make those conditions machine-auditable before any candidate result is promoted.

The project contains two different layers. The first is an earlier mathematical paper on statewise versus operator semantics at dimension two[3]. The present manuscript is a separate, scoped methods record. It preserves the published Paper I record and does not revise its claims. The second layer is the control and production protocol for larger finite-growth measurements. We close that layer for manuscript assembly at the bounded evidence boundary described below.

The global project verdict remains `FINAL_THEORY_OPEN`. In particular, we do not claim a complete finite occurrence-ON semantics lattice, a resolution of the full 955 or 721 profiles, candidate physical validity, a continuum limit, production spectral dimension, or a solver impossibility result.

The workflow is designed against four recurrent failure modes. First, a bounded fixture can be described as though it were a production sample. Second, two nominally independent checks can silently share the same implementation. Third, a timeout or enumeration cap can be rewritten as a negative theorem. Fourth, a valid temporary output can be promoted even though its certificate, source identity, or process cleanup failed. Each failure mode is represented by a machine-readable authorization field or a supervisor fixture below.

3 Machine-readable authorization model

Every stage has a versioned JSON result, a semantic SHA-256 digest, a report, and an explicit authorization boundary. A preflight may validate code paths and fixtures without authorizing production trajectories. The following invariants are required for the scoped manuscript:

1. bounded fixtures are labelled non-evidentiary;
2. primary and independent implementations agree before promotion;
3. candidate and control paths have separate identities and budgets;
4. approximate fallbacks are disabled unless a separate bias and mixing certificate exists;
5. a timeout, memory limit, or exact enumeration cap yields `RESOURCE_LIMIT_OPEN`, never a scientific negative result;
6. the published Paper I record is immutable and any later publication uses a new versioned archive.

The Phase-C artifact certifies the claim-boundary ledger, verdict grammar, frozen-artifact invariance, and preflight/budget separation. It adds no scientific verdict and keeps the solver gate closed.

4 Bounded exact controls

4.1 Finite sampler fixtures

Two distinct preflights are retained. A small- n sampler regression follows five fixed seeds, with 1,024 trajectories per seed, through cardinality three. Branch selection uses integer tickets over exact rational weights. The five terminal distributions pass their declared diagnostic comparison with the exact distribution. This comparison is a regression test; its six-sigma window is not a scientific acceptance threshold.

A later deterministic replay preflight checks two labelled implementation profiles, two seeds, two trajectory indices, and target sizes 1 through 5, giving 40 fixture trajectories. The finite catalogue reproduces the exact unlabelled level counts

$$1, 1, 2, 5, 16, 63$$

through cardinality five. Primary and independent relation and trajectory digests agree, a repeated primary fixture is deterministic, and both paths fail closed at their exact down-set caps. The presence of a candidate-labelled fixture validates only the frozen code path and identity boundary. It is not candidate evidence and does not authorize candidate sampling.

4.2 Four-dimensional positive control

The Minkowski control constructs fixed-cardinality Alexandrov-interval relations at bounded sizes 1, 4, 8, and 13. Eight primary/replay fixtures agree on relation digests, preserve cardinality, and pass the declared four-dimensional anchor check. This is an implementation preflight. The observable pipeline, production seed set, and independent observable digest are not promoted by this manuscript.

The positive control is deliberately separated from the BDG action proposal. No BDG coefficient fit, nonlocal scale fit, action-weighted ensemble, or candidate-derived calibration is performed.

4.3 Bounded cost observations

A three-case cost preflight completes one $N = 27$ random-growth fixture and one $N = 27$ Minkowski fixture. The planned $N = 60$ random-growth case returns `RESOURCE_LIMIT_OPEN` at the exact down-set cap of 4,096 after a bounded attempt. No cap increase or approximate fallback is applied. Wall times are environment-specific observations and are therefore kept in the machine artifact rather than treated as portable performance claims.

5 Independent replay and runtime safety

The replay contract fixes the ticket wire format, branch ordering, relation digest, and trajectory digest. The runtime layer rejects oversized tickets, excessive rejection attempts, exact down-set cap violations, and cyclic height dynamic programs. Hard-supervisor fixtures exercise wall-time limits, process-tree termination, memory and disk caps, invalid-digest rejection, atomic promotion, retry prohibition, and failed-artifact non-promotion.

The supervisor test matrix has five modes: success, invalid certificate, process-tree timeout, memory limit, and disk limit. Only the successful, digest-valid fixture is promoted. Every failed or resource-limited fixture is left unpromoted, each process tree is reaped, and no automatic retry occurs.

These checks establish the declared runtime contracts. They do not imply that the production sampler has been connected to the supervisor or that a production trajectory has been generated. The scoped closure records zero new control trajectories, zero candidate trajectories, and zero solver calls.

6 Exact scaling boundary

For the uniform random-growth control, every branch has weight one. The proposed exact route is an ideal-count recurrence over a maximal vertex v :

$$I(S) = I(S \setminus \{v\}) + I(S \setminus (\{v\} \cup \text{Pred}_S(v))), \quad I(\emptyset) = 1.$$

This recurrence is recorded as a design only; it is not implemented in the present closure.

The preceding exact scaling review uses the existing enumerating route under a cap of 4,096 down-sets. Both implementations complete 38 successful growth steps and produce the same cardinality-38 relation and trajectory digest. The last source for which all down-sets are enumerated is $n = 37$, with exact count 3,791. When the cardinality-38 relation is used as the next source, both implementations detect a lower bound of 4,097 down-sets and stop before drawing the next transition. Thus the recorded boundary is an attempted step *from* source size $n = 38$, not a failure to construct the size-38 relation.

This boundary supports the statement that the current exact enumerating path does not reach the planned $N = 60$ production scale under its declared cap. It does not support any of the following stronger statements: that no algorithm can scale, that the control has a physical continuum limit, or that a candidate is ruled out.

7 Candidate exclusion boundary

The active candidate profile is `causal_information_v2_sparse_kraus`. Its frozen weight has the form

$$w(P) = \left(\frac{2}{3}\right)^{\Delta\text{links}(P)} \left(\frac{3}{2}\right)^{\Delta\text{diamonds}(P)} \left(\frac{4}{5}\right)^{|P|}.$$

Unlike the uniform control, this is a nonlocal set function of the precursor. The repository contains no certified finite reduced-state recurrence or factorisation theorem that would permit exact weighted ticket selection without enumerating the relevant states. The uniform-control recurrence is therefore not reused for the candidate.

Candidate access is excluded from the scoped manuscript. This is a protocol decision, not a candidate obstruction or a negative physical result. The bounded candidate-labelled regression fixtures described above do not change this boundary: no candidate production observable, continuum estimate, or acceptance threshold is formed.

8 Evidence ledger and reproducibility

The closure packet binds the manuscript to the following existing artifacts:

Artifact	Role
phase-C completion	claim boundaries, verdict grammar, budget separation
small- n sampler preflight	exact-distribution regression at five fixed seeds
control sampler preflight	40 deterministic profile/replay fixtures
control implementation preflight	eight Minkowski positive-control fixtures
control cost/scaling review	exact cap boundary and cost observation
exact scaling design	uniform-DP proposal and candidate factorisation boundary
hard-supervisor preflight	fail-closed runtime fixture contract
candidate identity audit	candidate/control identity separation

Table 1: Digest-bound evidence chain for the scoped manuscript.

The repository contains the corresponding JSON results, reports, source modules, and tests. Each JSON artifact carries a semantic SHA-256 digest; the publication supplement additionally records raw SHA-256 hashes, byte counts, the source commit, and the standalone PDF hash. The supplement is therefore an authenticated navigation and reproduction layer, not a replacement for the machine-readable ledgers.

8.1 Reproduction protocol

A clean reproduction has three levels. First, verify the archived hashes and the source-commit binding without executing research code. Second, run the focused Phase-B and Phase-C tests, followed by lint and static type checks. Third, if a fresh PDF is required, build the TeX source in the digest-pinned, network-disabled container recorded in the PDF build report. Reproduction of existing fixtures does not reopen candidate or production authorization.

The upload layout consists of one directly previewable PDF and one deterministic compressed supplement. The PDF is not duplicated inside the supplement; the supplement binds it by SHA-256. Raw third-party papers, extracted reference texts, credentials, caches, and unbounded solver outputs are excluded.

9 Unresolved work and reopening rule

The following remain outside the present paper: production spectral-window and statistical freezes, $N=60$ control training/holdout, an executable BDG ensemble, candidate weighted production sampling, and full 955/721 classification. A future reopening requires a new owner-approved scope, a versioned budget, independent replay, and (for candidate access) a certified weighted-state factorisation. The existing Paper I Zenodo record must not be mutated.

10 Conclusion

The project now has a finite, auditable methods boundary. Bounded controls, replay, resource guards, and candidate exclusion are machine-recorded, while the exact scaling stop is reported as a resource boundary rather than a scientific no-go. This supports a scoped control-and-methods manuscript. It does not close the physical theory question, which remains `FINAL_THEORY_OPEN`.

Data and code availability

This preprint is accompanied by a deterministic supplement containing its TeX source, bibliography, reproduction instructions, selected machine-readable artifacts, tests, release metadata, and source-commit manifest. The repository is available at <https://github.com/To-marigi/universe-theory-lab>; the exact source commit for this version is recorded inside the supplement. Paper I v0.4.2 remains available at <https://zenodo.org/records/21861533>; that record is not modified by this preprint.

License

The manuscript and its original textual material are released under the Creative Commons Attribution 4.0 International license. Software included in the supplement remains under the repository’s MIT license. Third-party works are not redistributed in the supplement.

AI and human-author disclosure

AI tools assisted with bounded implementation inspection, test scaffolding, artifact comparison, and prose drafting. They are not authors or independent proof authorities. The human author retains responsibility for scope, evidence, interpretation, and release approval.

References

- [1] Lisa Glaser, Denjoe O’Connor, and Sumati Surya. Finite size scaling in 2d causal set quantum gravity. *Classical and Quantum Gravity*, 35:045006, 2018.
- [2] Stan Gudder. Causal set approach to discrete quantum gravity, 2012.
- [3] Kenichi Osaki. Statewise versus operator bell causality in finite quantum sequential growth: exact separation and recovery at dimension two, 2026.
- [4] D. P. Rideout and R. D. Sorkin. A classical sequential growth dynamics for causal sets. *Physical Review D*, 61:024002, 2000.
- [5] Ritesh Srivastava and Sumati Surya. Implementing bell causality in quantum sequential growth, 2026. Version 1.
- [6] Sumati Surya and Stav Zalel. A criterion for covariance in complex sequential growth models. *Classical and Quantum Gravity*, 37:195030, 2020.